

Lexicon

AI-Powered Document Question Answering System

Prompt engineering, embeddings and Retrieval-Augmented Generation

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How the system works

1 Chunk

Split each PDF, TXT or MD file into 500-character chunks with 50 overlap.

2 Embed

all-MiniLM-L6-v2 turns every chunk into a 384-number vector.

3 Store

Vectors and text go into ChromaDB using cosine distance.

4 Retrieve

The question is embedded and the 3 nearest chunks are returned.

5 Generate

Groq answers from those chunks with one of three prompt styles.

Every answer cites its source files, and the top-3 chunks are shown with similarity scores so each answer can be checked.

Three prompting techniques

Zero-shot

Context and question only.
No examples, no role.

Answer the question using the context below.

Few-shot

Three worked examples set the style, including one that says the answer is not in the documents.

*Q: What does ChromaDB do?
A: It stores embeddings and finds the most similar ones.*

Role-based

A senior technical document analyst with strict rules: context only, direct answer first, cite sources.

Start with a one-sentence answer. Mention source files in brackets.

Semantic search with embeddings

- Loads 5 documents (PDF, TXT, Markdown)
- Chunks of 500 characters, 50 overlap
- Sentence-transformer embeddings, 384 dimensions
- Stored in ChromaDB with cosine distance
- Question embedded the same way
- Top 3 chunks returned with similarity and distance

Reading the scores

distance = 1 - similarity

Similarity near 1 means the chunk points the same way as the question. Near 0 means unrelated. Lower distance means a better match, so results are sorted by distance.

Chunk overlap keeps boundary sentences findable

Semantic vs keyword search

	Keyword search	Semantic search
Matches on	Exact words in the query	Meaning, through embeddings
Synonyms	Missed unless the word appears	Found: "fix a car" matches "repairing an automobile"
Ranking	Count of shared words	Closeness of vectors (cosine)
Best for	Names, codes, exact phrases	Natural-language questions
Weakness	Fails when wording differs	Can return related but not exact text

Lexicon shows both results side by side for any query in the Semantic vs keyword tab.

Comparing the three prompts

How we test

- Same 5 questions for every technique
- Same retrieved chunks for fairness
- An LLM judge scores 1 to 10
- Criteria: accuracy, clarity, relevance
- Average per technique decides the winner

Results from one run

Role-based: 9.66

Zero-shot: 9.60

Few-shot: 9.54

Averages of accuracy, clarity and relevance (out of 10). The gaps are small, so role-based has a slight edge, not a decisive one.

Deployment and takeaways

Deployment

- Vercel: FastAPI backend and frontend
- Firebase Hosting: optional frontend
- ONNX embeddings keep the bundle small
- Groq key stored as an environment variable

Takeaways

- Retrieval grounds answers in real documents
- Prompt design changes accuracy and clarity
- Semantic search finds meaning, not just words
- Next: persistent storage and more file types